LECTURE 04 - EXPONENTIAL MAPS AND COVERINGS

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1. Exponential Maps

Let G, H, etc. be a Lie groups and $\mathfrak{g}, \mathfrak{h}$, etc. be their Lie algebras. Recall the following Proposition summarized from Lecture Notes 03. As usual, a simply connected manifold is always assumed to be connected.

Proposition 1.1. (1) *If $\varphi : G \to H$ is a Lie group homomorphism, then $d\varphi$, the tangent map at e, is a Lie algebra homomorphism from $\mathfrak{g}$ to $\mathfrak{h}$. When G is connected, the map $\varphi \mapsto d\varphi$ is injective.*

(2) *Conversely, if G is simply connected, given a Lie algebra homomorphism $\psi : \mathfrak{g} \to \mathfrak{h}$, there exists a unique group homomorphism $\varphi : G \to H$ lifting ψ.*

Example 1.2. *Consider the character $\chi_s : \mathbb{R}^\times \to \mathbb{C}^\times, a \mapsto |a|^s$, where $s \in \mathbb{R}$. We learn in calculus that the derivative of χ_s at 1, which is s, measures the dilation of the one-dimensional tangent space $T_1\mathbb{R}^\times = \mathbb{R}$. What it means in Lie theory is that the character χ_s induces a Lie algebra homomorphism $\psi_s : \mathbb{R} \to \mathbb{C}, a \mapsto sa$. As the tangent space gives no information on the connected component of $\mathbb{R}^\times$ not containing 1, $d\chi_s$ does not determine χ_s uniquely. In fact, $\chi_s' : \mathbb{R}^\times \to \mathbb{C}^\times, a \mapsto |a|^s \mathrm{sgn}(a)$ is another character inducing the same Lie algebra homomorphism ψ_s. If we change $\mathbb{R}^\times$ to S^1, then we learn from HW 2 that not all Lie algebra homomorphism ψ_s can be lifted to a character of S^1. Only the Lie algebra homomorphism ψ_n with $n \in \mathbb{Z}$ can be lifted to a character $\chi_n : S^1 \to \mathbb{C}^\times, z \mapsto z^n$.*

Definition 1.3. *Let $X \in \mathfrak{g}$. The Lie algebra homomorphism $\mathbb{R} \to \mathfrak{g}$, $\frac{d}{dr} \mapsto X$ can be uniquely lifted to a Lie group homomorphism $\exp_X : \mathbb{R} \to G$. Then $\exp_X(t)$ is the unique solution of the ODE on the group G*

$$\frac{d}{dt}\sigma(t) = X_{\sigma(t)}$$

with the initial condition $\sigma(0) = e$. The ***exponential map*** *is defined to be*

$$\exp : \mathfrak{g} \to G, \quad X \mapsto \exp_X(1).$$

Remark 1.4. *An injective group homomorphism $\varphi : \mathbb{R} \to G$ is called a* ***one-parameter subgroup*** *of G.*

Theorem 1.5 ([FWar83, Theorem 3.31])**.** *Let $X \in \mathfrak{g}$, $t, t_1, t_2 \in \mathbb{R}$.*

(1) $\exp(tX) = \exp_X(t)$
(2) $\exp(t_1 + t_2)X = \exp(t_1X)\exp(t_2X)$
(3) $\exp(-tX) = (\exp(tX))^{-1}$
(4) $\exp : \mathfrak{g} \to G$ *is a local diffeomorphism at zero. Here we identify $\mathfrak{g} \simeq T_eG$ and equip $\mathfrak{g}$ with the standard Euclidean smooth structure.*

Proof. Let $s, t \in \mathbb{R}$. Consider $\varphi(t) := \exp_X(st)$ and $\phi(t) := \exp_{sX}(t)$. Then $\varphi(0) = \phi(0) = e$ and both satisfy the ODE

$$\frac{d}{dt}\varphi(t) = sX_{\varphi(t)}, \qquad \frac{d}{dt}\phi(t) = sX_{\phi(t)}.$$

Thus, the uniqueness of solutions of ODE implies that $\exp_X(st) = \exp_{sX}(t)$. Hence (1) follows. Then (2), (3) follow immediately from (1). For instance,

$$\exp(t_1 + t_2)X = \exp_X(t_1 + t_2) = \exp_X(t_1)\exp_X(t_2) = \exp(t_1X)\exp(t_2X).$$

To prove (4), we identify $\mathfrak{g}$ with the tangent space of $\mathfrak{g}$ at any point $X \in \mathfrak{g}$. Consider the smooth vector field defined on the product manifold $G \times \mathfrak{g}$,

$$Y_{(g,X)} := (X_g, 0) \in T_g(G) \times T_X(\mathfrak{g}).$$

Here we identify $\mathfrak{g}$ with the tangent space of $\mathfrak{g}$ at X. Then $\varphi(g, t) := (g\exp(tX), X)$ is the unique solution of the ODE

$$\frac{\partial}{\partial t}\sigma(g, t) = (X_{g\exp(tX)}, 0), \qquad \sigma(0) = (g, X).$$

For every fixed $g \in G$, such a solution is automatically smooth (as the vector field Y is smooth when g is fixed). Thus

$$\exp(X) = p_1(\sigma(e, t))$$

is smooth. Here p_1 is the projection to the first coordinate. Moreover, the above implies

$$d\exp(X_e) = dp_1 \circ (X_e, 0) = X_e.$$

Thus the differential of exp at identity is the identity map. This means exp must be a local diffeomorphism. □

Theorem 1.6 ([FWar83, Theorem 3.32])**.** *Let $\varphi : H \to G$ be a homomorphism of Lie groups. Then the following diagram commutes.*

$$\begin{tikzcd} H \arrow[r, "\varphi"] & G \\ \mathfrak{h} \arrow[u, "\exp"] \arrow[r, "d\varphi"] & \mathfrak{g} \arrow[u, "\exp"'] \end{tikzcd}$$

Proof. Let $X \in \mathfrak{g}$. Both $t \mapsto \varphi(\exp(tX))$ and $t \mapsto \exp(td\varphi(X))$ are one-parameter subgroups of G whose tangent vector at e is equal to $d\varphi(X)$. By uniqueness,

$$\varphi(\exp(tX)) = \exp(td\varphi(X).$$

Thus, the theorem follows. □

Corollary 1.7. *Let H be a Lie subgroup of G, not necessarily closed. Then the Lie algebra*

$$\mathfrak{h} = \{X \in \mathfrak{g} \mid \exp tX \in H, \quad \text{for all } t \in \mathbb{R}\}.$$

Proof. This follows directly from Theorem 1.6 (by setting φ to be the inclusion map). □

Corollary 1.8 ([FWar83, Theorem 3.43])**.** *Suppose we have a Lie group homomorphism $\varphi : G \to H$. Then $\ker\varphi$ is a closed Lie subgroup with Lie algebra $\ker d\varphi$.*

Proof. Obviously $\ker\varphi$ is a closed subgroup, hence it inherits a Lie group structure. Then Corollary 1.7 implies that the Lie algebra of $\ker\varphi$ is equal to

$$\{X \in \mathfrak{g} \mid \exp tX \in \ker\varphi, \quad \forall t \in \mathbb{R}\}.$$

By Theorem 1.6, this set is exactly

$$\{X \in \mathfrak{g} \mid \exp(d\varphi(tX)) = e, \quad \forall t \in \mathbb{R}\}.$$

As exp is a local diffeomorphism at 0, by enforcing t to be sufficiently small, we see that the above set is exactly $\ker d\varphi$. This finishes the proof. □

Example 1.9. *Let $F = \mathbb{R}$ or $\mathbb{C}$. Consider the space $\mathrm{Mat}_n(F)$ equipped with the operator norm. Then for each $X \in \mathrm{Mat}_n(F)$, the series*

$$e^X := \sum_{k=0}^{\infty} \frac{1}{k!} X^k$$

converges absolutely and uniformly on compacta in $\mathrm{Mat}_n(F)$. Hence $t \mapsto e^{tX}$ is a Lie group homomorphism from $\mathbb{R} \mapsto \mathrm{GL}_n(F)$, i.e. a one-parameter subgroup of $\mathrm{GL}_n(F)$. Hence by differentiating termwise,

$$\frac{d}{dt}\Big|_{t=0} e^{tX} = X.$$

This shows that the tangent vector of e^{tX} at identity is equal to X. Thus for $\mathrm{GL}_n(F)$, the exponential map is exactly the exponential of matrices. In view of Theorem 1.6, the exponential map for every linear Lie group (i.e. Lie subgroup of $\mathrm{GL}_n(F)$), exponential map is given by exponential of matrices.

Example 1.10. *In this example, we discuss the explicit formula of exponential map for $\mathrm{SL}_2(\mathbb{R})$. Let $A = \begin{pmatrix} a & b \\ c & -a \end{pmatrix} \in \mathfrak{sl}_2(\mathbb{R})$. Then $A^2 = (a^2 + bc)\mathrm{I}_2 = -\det(A)\mathrm{I}_2$.*

If $a^2 + bc = 0$, then $e^A = \mathrm{I}_2 + A$.

If $a^2 + bc < 0$, we set $\omega := \sqrt{-(a^2 + bc)}$. Then $A^2 = -\omega^2\mathrm{I}_2$. Plugging into the infinite series of e^A, we see that

$$\begin{aligned} e^A &= \left(\mathrm{I}_2 - \frac{1}{2!}\omega^2 + \frac{\omega^4}{4!} - \cdots\right)\mathrm{I}_2 + \frac{1}{\omega}\left(\omega - \frac{1}{3!}\omega^3 + \frac{1}{5!}\omega^5 - \cdots\right)A \\ &= \cos\omega\mathrm{I}_2 + \frac{\sin\omega}{\omega}A. \end{aligned}$$

If $a^2+bc>0$, we set $\omega := \sqrt{a^2+bc}$. Then $A^2=\omega^2 \mathrm{I}_2$. Plugging into the infinite series of e^A, we see that

$$\begin{aligned} e^A &= \left(\mathrm{I}_2 + \frac{1}{2!}\omega^2 + \frac{\omega^4}{4!} + \cdots\right)\mathrm{I}_2 + \frac{1}{\omega}\left(\omega + \frac{1}{3!}\omega^3 + \frac{1}{5!}\omega^5 + \cdots\right)A \\ &= \cosh\omega \mathrm{I}_2 + \frac{\sinh\omega}{\omega}A. \end{aligned}$$

Here $\cosh\omega = \frac{1}{2}(e^\omega + e^{-\omega})$, $\sinh\omega = \frac{1}{2}(e^\omega - e^{-\omega})$.

Example 1.11. *As e^z, $\sin z$ etc. are also well-defined for $z\in\mathbb{C}$, the computation in Example 1.10 works via a word by word repetition. In a compact form, we set $\omega\in\mathbb{C}$ such that $\omega^2 = -\det A$. Then the exponential map for $\mathrm{SL}_2(\mathbb{C})$ is given explicitly by*

$$e^A = \cosh\omega \mathrm{I}_2 + \frac{\sinh\omega}{\omega}A.$$

Example 1.12. *In this example, we discuss the explicit formula of exponential map for* $\mathrm{SO}(3)$. *Let*

$$A = \begin{pmatrix} & a & b \\ -a & & c \\ -b & -c & \end{pmatrix} \in \mathfrak{so}(3).$$

Then

$$A^2 = \begin{pmatrix} -a^2-b^2 & -bc & ac \\ -bc & -a^2-c^2 & -ab \\ ac & -ab & -b^2-c^2 \end{pmatrix}.$$

Set $\theta = \sqrt{a^2+b^2+c^2}$. Then $A^3 = -\theta^2 A$. By a simple computation, it is easy to show that

$$e^A = \mathrm{I}_3 + \frac{\sin\theta}{\theta}A + \frac{1-\cos\theta}{\theta^2}A^2.$$

Example 1.13. *Let $G\le \mathrm{GL}_n(\mathbb{C})$ be a linear Lie group. Then for any $X\in\mathfrak{g}\le\mathfrak{gl}_n(\mathbb{C})$,* $\det e^X = e^{\mathrm{Tr}X}$. *This means that $d(\det) = \mathrm{Tr}$.*

Indeed, let $X = PJP^{-1}$, where J is the Jordan canonical form. Then $\det e^X = \det(Pe^JP^{-1}) = \det e^J$ and $e^{\mathrm{Tr}X} = e^{\mathrm{Tr}J}$. The identity $\det e^J = e^{\mathrm{Tr}J}$ is now obvious (as J is upper triangular). The identity $\det e^X = e^{\mathrm{Tr}X}$ can be translated to the following commutative diagram

$$\begin{tikzcd} G \arrow[r, "\det"] & \mathbb{C}^\times \\ \mathfrak{g} \arrow[u, "\exp"] \arrow[r, "\mathrm{Tr}"] & \mathbb{C} \arrow[u, "\exp"'] \end{tikzcd}.$$

This means that $d(\det) = \mathrm{Tr}$.

Example 1.14. *In this example, we discuss the inverse of* exp *for linear Lie groups. Recall that*

$$\ln(1+z) = \sum_{j=1}^{+\infty} \frac{(-1)^{j+1}}{j} z^j$$

converges absolutely in the open disc $|z|<1$. It is the inverse e^z in the sense that

$$e^{lnz} = z \quad \text{if } |z-1|<1$$

and

$$\ln(e^z) = z \quad \text{if } |e^z - 1| < 1.$$

This implies that matrix logarithm

$$\ln A := \sum_{j=1}^{+\infty} \frac{(-1)^{j+1}}{j}(A-I)^j, \quad \|A - I\| < 1$$

is the inverse of matrix exponential (in the range of convergence).

2. A Few Words on $\mathrm{Aut}(G)$, $\mathrm{Aut}(\mathfrak{g})$ and $\mathrm{Der}(\mathfrak{g})$

Let G be a Lie group, assumed to be connected for simplicity. For every $\varphi \in \mathrm{Aut}(G)$ (the abstract group of Lie group automorphisms), by Proposition 1.1, $d\varphi \in \mathrm{Aut}(\mathfrak{g})$ (the group of Lie algebra automorphisms). Equip $\mathfrak{g}$ the usual Euclidean vector space topology, then $\mathrm{GL}(\mathfrak{g})$ is a Lie group whose topology is inherited from $\mathrm{End}_{\mathbb{R}}(\mathfrak{g})$. As $\mathrm{End}_{\mathbb{R}}(\mathfrak{g})$ is finite-dimensional, the Euclidean topology is the same as the operator norm topology, and it is the same as the strong topology (standard functional analysis). This means that for a sequence $\varphi_j \in \mathrm{End}_{\mathbb{R}}(\mathfrak{g})$ to converge to $\varphi \in \mathrm{End}_{\mathbb{R}}(\mathfrak{g})$, we only need to check

$$\lim_j \varphi_j(X) = \varphi(X) \qquad \forall X \in \mathfrak{g}.$$

The clearly implies that the set of all endomorphisms of $\mathfrak{g}$ is closed in $\mathrm{End}_{\mathbb{R}}(\mathfrak{g})$. Hence under the relative topology,

Lemma 2.1. $\mathrm{Aut}(\mathfrak{g})$ *is a closed Lie subgroup of* $\mathrm{GL}(\mathfrak{g})$.

Remark 2.2. $\mathrm{Aut}(\mathfrak{g})$ *means* $\mathrm{Aut}_{\mathbb{R}}(\mathfrak{g})$*, the group of* $\mathbb{R}$*-linear automorphisms of* $\mathfrak{g}$*. If* $\mathfrak{g}$ *is further assumed to be a complex Lie algebra, we may have to emphasize the field, meaning that we need to distinguish* $\mathrm{Aut}_{\mathbb{R}}(\mathfrak{g})$ *against* $\mathrm{Aut}_{\mathbb{C}}(\mathfrak{g})$.

Proposition 2.3. *Let* $\mathfrak{g}$ *be a finite-dimensional Lie algebra over* F *(*$F = \mathbb{R}$ *or* $\mathbb{C}$*). Recall that*

$$\mathrm{Der}_F(\mathfrak{g}) := \{D \in \mathrm{End}_F(\mathfrak{g}) \mid D[X,Y] = [DX,Y] + [X,DY]\}.$$

Then the Lie algebra of $\mathrm{Aut}_F(\mathfrak{g})$ *(the group of* F*-linear automorphisms of* $\mathfrak{g}$*) is* $\mathrm{Der}_F(\mathfrak{g})$.

Proof. Homework. □

Now consider the differential map $d : \mathrm{Aut}(G) \to \mathrm{Aut}(\mathfrak{g})$, again, Proposition 1.1 implies that this differential map is injective. Thus, one may wish to regard $\mathrm{Aut}(G)$ as a Lie subgroup of $\mathrm{Aut}(\mathfrak{g})$. Yet what is the topology on $\mathrm{Aut}(G)$? Hmm..... we do not want to spend too much time on functional analysis, so let us just dive into the answer.

Theorem 2.4 (Hochschild, 1952)**.** *Let* G *be a connected Lie group. Naturally equip* $\mathrm{Aut}(G)$ *with the compact-open topology. Then the differential map*

$$d : \mathrm{Aut}(G) \to \mathrm{Aut}(\mathfrak{g}), f \mapsto df$$

is a homeomorphism onto the image, i.e. $\mathrm{Aut}(G)$ *is a closed Lie subgroup of* $\mathrm{Aut}(\mathfrak{g})$.

Remark 2.5. *For practical reasons, we do not consider Lie groups with too many connected components. To be precise, recall that in Lecture 03 Theorem 1.19, we mentioned that* G/G^0 *is a group isomorphic to* $\pi_0(G)$*. When* G/G_0 *is finitely generated, same paper of Hochschild also proves that* $\mathrm{Aut}(G)$ *is a Lie group.*

3. Semi-direct Product of Lie groups and Lie algebras

Recall that for any Lie algebra $\mathfrak{g}$ over a field F, the space of **derivations**

$$\mathrm{Der}_F(\mathfrak{g}) := \{D \in \mathrm{End}_F(\mathfrak{g}) \,|\, D \text{ satisfies the Leibniz rule } D[X,Y] = [DX,Y] + [X,DY]\}$$

forms a Lie algebra over F.

Definition 3.1. *Given two Lie algebras $\mathfrak{a}, \mathfrak{b}$ over F and a homomorphism between Lie algebras*

$$\pi : \mathfrak{b} \to \mathrm{Der}_F(\mathfrak{a}),$$

we can equip the vector space $\mathfrak{g} := \mathfrak{b} \oplus \mathfrak{a}$ an F-Lie algebra structure by preserving the Lie bracket relations within $\mathfrak{a}$ and $\mathfrak{b}$ and defining

$$[B, A] := \pi(B)(A)$$

for all $A \in \mathfrak{a}$ and $B \in \mathfrak{b}$. Explicitly,

$$[(B_1, A_1), (B_2, A_2)] := ([B_1, B_2], \pi(B_1)A_2 - \pi(B_2)A_1 + [A_1, A_2]).$$

*Then we call $\mathfrak{g}$ a **semidirect product** of $\mathfrak{a}$ and $\mathfrak{b}$. In this way, $\mathfrak{a}$ (resp. $\mathfrak{b}$) is realized as an ideal (resp. a subalgebra) of $\mathfrak{g}$. We write*

$$\mathfrak{g} = \mathfrak{b} \rtimes_\pi \mathfrak{a}.$$

If the action π is trivial, then we just say $\mathfrak{g}$ is a direct sum of $\mathfrak{a}$ and $\mathfrak{b}$.

Example 3.2. *Consider $\mathfrak{b} := \mathfrak{o}(n)$, $\mathfrak{a} := \mathbb{R}^n$ (regarded as an abelian Lie algebra). Then $\mathrm{Der}_{\mathbb{R}}(\mathfrak{a}) = \mathrm{End}_{\mathbb{R}}(\mathfrak{a})$. Now we take $\pi : \mathfrak{b} \to \mathrm{Der}_{\mathbb{R}}(\mathfrak{a})$ to be the natural inclusion, this leads to a semidirect product*

$$\mathfrak{g} := \mathfrak{b} \rtimes_\pi \mathfrak{a} = \mathfrak{o}(n) \rtimes_\pi \mathbb{R}^n.$$

$\mathfrak{g}$ is called the Lie algebra of the rigid motion group on $\mathbb{R}^n$. To realize $\mathfrak{g}$ in terms of matrices, we realize

$$\mathfrak{g} = \left\{ \begin{pmatrix} A & v \\ 0 & 0 \end{pmatrix} \,\middle|\, A \in \mathfrak{o}(n), v \in \mathbb{R}^n \right\}.$$

Then the action of $\mathfrak{o}(n)$ on $\mathbb{R}^n$ is realized as the Lie bracket on $\mathfrak{g}$:

$$[\begin{pmatrix} A & 0 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & v \\ 0 & 0 \end{pmatrix}] = \begin{pmatrix} 0 & Av \\ 0 & 0 \end{pmatrix}$$

Let G and H be Lie groups. For practical reasons, we only work under the condition that $\pi_0(G)$ is finitely generated so that Hochschild's theorem holds.

Definition 3.3. *Given a Lie group homomorphism $\tau : G \to \mathrm{Aut}(H)$, we can define a Lie group structure on the product manifold $G \times H$. The group multiplication is given by*

$$(g_1, h_1)(g_2, h_2) = (g_1 g_2, h_1 \cdot \tau(g_1) h_2).$$

In this way, H is a normal subgroup of the semidirect product $G \rtimes_\varphi H$ and G is a closed subgroup.

Given a semidirect product $G \rtimes_\varphi H$, by considering

$$G \xrightarrow{\varphi} \mathrm{Aut}(H) \xrightarrow{d} \mathrm{Aut}(\mathfrak{h}),$$

we get a commutative diagram

$$\begin{tikzcd} G \arrow[r, "\varphi"] & \mathrm{Aut}(\mathfrak{h}) \\ \mathfrak{g} \arrow[u, "\exp"] \arrow[r, "d\varphi"] & \mathrm{Der}(\mathfrak{h}). \arrow[u, "\exp"'] \end{tikzcd}$$

Here for simplicity, we identify φ with $d \circ \varphi$. Thus,

Proposition 3.4. *The Lie algebra of $G \rtimes_\varphi H$ is $\mathfrak{g} \rtimes_{d\varphi} \mathfrak{h}$. Conversely, when G and H are simply connected, given any Lie algebra homomorphism $\psi : \mathfrak{g} \to \mathrm{Der}(\mathfrak{h})$, there exists a unique semidirect product $G \rtimes_\varphi H$ lifting $\mathfrak{g} \rtimes_{d\varphi} \mathfrak{h}$.*

Proof. The first statement is trivial. As for the second, note that as G is simply connected, we can first lift ψ to a Lie group homomorphism $\varphi : G \to \mathrm{Aut}(\mathfrak{h})$. When H is simply connected, $\mathrm{Aut}(H) = \mathrm{Aut}(\mathfrak{h})$ in view of Proposition 1.1. □

Example 3.5. *Consider $G := \mathrm{O}(n)$, $H = \mathbb{R}^n$. Take $\varphi : G \to \mathrm{Aut}(\mathbb{R}^n) = \mathrm{GL}_n(\mathbb{R})$ to be the natural inclusion, this leads to a semidirect product*

$$\mathrm{O}(n) \rtimes \mathbb{R}^n.$$

Such group is called the rigid motion group on $\mathbb{R}^n$. To realize $\mathfrak{g}$ in terms of matrices, we realize

$$\mathrm{O}(n) \rtimes \mathbb{R}^n = \left\{ \begin{pmatrix} A & v \\ 0 & 1 \end{pmatrix} \,\middle|\, A \in \mathrm{O}(n), v \in \mathbb{R}^n \right\}.$$

4. Surjectivity and Injectivity of Exponential Maps

We always assume that G is connected in this section.

Definition 4.1. *A Lie group is said to be **solvable** (resp. **nilpotent**) if it is connected and its Lie algebra is **solvable** (resp. **nilpotent**).*

Remark 4.2. *By definition, abelian groups are always nilpotent. (I am not going to use the term unipotent groups here. Unipotent groups work better in the algebraic setting.) The abelian group S^1 is considered to be nilpotent, yet it is not an algebraic group (it is the real points of an algebraic group).*

Theorem 4.3 ([Kna02, Theorem 1.127, Corollary 134])**.** *Let N be a simply connected nilpotent Lie group with Lie algebra $\mathfrak{n}$. Then $\exp : \mathfrak{n} \to N$ is a diffeomorphism. As a consequence, any Lie subgroup of N is simply connected and closed.*

Idea of proof. As $\mathfrak{n}$ is nilpotent, we get an decreasing sequence

$$\mathfrak{n} \supset \mathfrak{n}_1 \supset \mathfrak{n}_2 \supset \cdots \supset \mathfrak{n}_{n-1} \supset \mathfrak{n}_n = 0$$

such that $\mathfrak{n}_j$ is a codimensional one space in $\mathfrak{n}_{j-1}$. As $\mathfrak{n}_{n-1}$ contains in the center of $\mathfrak{n}$, the corresponding Lie subgroup N_{n-1} contains in the center of N, hence normal. Now we get a simply connected nilpotent Lie group N/N_{n-1} with Lie algebra $\mathfrak{n}/\mathfrak{n}_{n-1}$. This indicates a proof on the theorem based on the induction on the dimension of N. Details are omitted. □

Surely, if one wants the exponential map to be surjective, then the Lie group G must be assumed to be connected. If one wants the exponential map to be injective, then naturally the group G should be assumed to be simply connected (think about the example S^1). But is it enough?

Example 4.4. *The rigid motion group* $E(2) := \mathrm{SO}(2) \rtimes \mathbb{R}^2$ *is connected and solvable. Let* $\widetilde{E}(2)$ *be its universal covering group (details will be discussed in the next section). Then the exponential map of* $\widetilde{G}$ *is neither injective nor surjective. (Details will be filled out as a Homework problem.)*

The following theorem proved by Diximier and Saito provide a complete characterization on this issue. (To be precise, Saito proved a result closely related, yet not included below. I guess it is still decent to put his name here.) I sketch a proof below for my personal convenience, yet students are not required to understand it.

Theorem 4.5 (Diximier, Saito (1957))**.** *The following are equivalent*

(1) exp *is injective;*
(2) exp *is bijective;*
(3) exp *is a diffeomorphism;*
(4) G *is solvable, simply connected, and* $\mathfrak{g}$ *does not admit* $\mathfrak{e}(2)$ *as a subalgebra of any quotient algebra.*

Proof. If G is assumed to be complex and simply connected, then the equivalence of (1),(2),(3) is exactly [**?**, Corollary of Theorem 1]. In this case, G is not only solvable, it is in fact nilpotent, hence (4) also follows ($\mathfrak{e}(2)$ is not nilpotent). If G is only assumed to be a simply connected Lie group, then the equivalence of (2),(3),(4) is exactly [**?**, Theorem 3].

To finish the proof, we note that the injectivity of the exponential map implies that G has no subgroup isomorphic to S^1. This means that the maximal compact subgroup of G must be trivial. (Why? Because any connected nontrivial compact Lie group admits a maximal torus, which is homeomorphic to some $(S^1)^n$! This is called the maximal torus theorem and will be covered later in the course.) The most general version of Malcev-Iwasawa theorem says that any connected Lie group must be homotopic to its maximal compact subgroup. In our case, G is homotopic to the single point, i.e. G is contractible.

Apply the Levi-decomposition on G. G is a semidirect product of a contractible solvable Lie group and a contractible semisimple Lie group. Again, for the semisimple factor, it is homotopic to its maximal compact subgroup. Hence the semisimple factor is contractible if and only if its maximal compact subgroup is trivial. This only happens for the universal cover of $\mathrm{SL}_2(\mathbb{R})$. Yet for $\widetilde{\mathrm{SL}_2(\mathbb{R})}$, the exponential map is not injective. So G must be a simply connected solvable group.

To finish the proof, one has to check (1) implies the second condition of (4). Since (1) implies that G is simply connected and solvable, which is the assumption taken by Diximier, then (1) $\Rightarrow$ (4) is implicitly proved in his paper. □

In this course, we will (hopefully) prove that exponential maps are surjective for compact Lie groups. But for classical compact groups, we can proceed as follows. Let us first prove that

Proposition 4.6. *The exponential maps for connected abelian groups are surjective.*

We need the following Lemma that will be proved in a more general setup in the next Lecture.

Lemma 4.7. *Assume that* G *is connected. Then* G *is abelian if and only if* $\mathfrak{g}$ *is abelian.*

Also recall the following Lemma proved in Lecture 03.

Lemma 4.8 ([Sep07, Theorem 1.15])**.** *Let U be a connected open neighbourhood of e. Then $G^0 = \bigcup_{n=1}^{\infty} U^n$.*

Lemma 4.9. *For $X, Y \in \mathfrak{g}$, $[X, Y] = 0$ if and only if $\exp(sX)\exp(tY) = \exp(tY)\exp(sX)$ for all $s, t \in \mathbb{R}$. In this situation, $\exp(sX)\exp(tY) = \exp(sX + tY)$*

Proof. Notice that $[X, Y] = 0$ if and only if $\mathfrak{h} := \mathbb{R} \cdot X + \mathbb{R} \cdot Y$ is an abelian Lie subalgebra of $\mathfrak{g}$. Moreover, $\exp(sX)\exp(tY) = \exp(tY)\exp(sX)$ for all $s, t \in \mathbb{R}$ if and only if $\exp(tX)$ and $\exp(sY)$ generate an abelian Lie subgroup H of G. Then the lemma follows easily from Lemma 4.7. □

Corollary 4.10. *The exponential maps for connected abelian groups are surjective.*

Proof. Let G be connected and abelian. Take a small neighbourhood U of $0 \in \mathfrak{g}$ such that $\exp|_U$ is a diffeomorphism onto its image. Set $V = \exp U$. Then

$$G = \bigcup_{n=1}^{\infty} V^n.$$

By Lemma 4.9, $V^n = (\exp U)^n \subseteq \exp(nU) \subseteq \exp \mathfrak{g}$. Thus $G \subseteq \exp \mathfrak{g}$. □

Theorem 4.11. *The exponential map is surjective for $G = \mathrm{U}(n), \mathrm{SO}(n), \mathrm{Sp}(2n)$.*

Proof. Let $g \in G$ and $X \in \mathfrak{g}$. Note that if $g = \exp X$, then for any $P \in G$, $PgP^{-1} = P(\exp X)P^{-1} = \exp(PXP^{-1})$. Then the theorem follows directly from Corollary 4.10, Theorem A.3 (for $\mathrm{SO}(n)$), Theorem A.2 (for $\mathrm{U}(n)$), Theorem A.4 (for $\mathrm{Sp}(2n)$). □

5. Covering, Center and π_1

Let G, H, etc. be a Lie groups and $\mathfrak{g}, \mathfrak{h}$, etc. be their Lie algebras.

In this section, all groups are assumed to be connected. Recall the notion of covering map.

Definition 5.1. *Let $\pi : \widetilde{G} \to G$ be a smooth map between connected smooth manifolds. Then π is said to be a* ***covering map*** *if it is surjective; and for each $g \in G$, there exists a neighbourhood U of g such that π is a diffeomorphism when restricted to any connected component of $\pi^{-1}(U)$. When $\widetilde{G}$ is simply connected, then we say $\widetilde{G}$ is a universal covering manifold of G.*

Universal covering manifold always exists and is unique up to diffeomorphism. The key point is to equip it a Lie group structure such that the covering map π is a homomorphism.

Theorem 5.2 ([FWar83, Theorem 3.25])**.** *The universal covering $\widetilde{G}$ can be equipped with a Lie group structure such that the covering map $\pi : \widetilde{G} \to G$ is a Lie group homomorphism.*

Proof. The proof is of no use in this course. We omit it. □

Corollary 5.3 ([FWar83, Proposition 3.26])**.** *Let $\varphi : G \to H$ be a homomorphism between connected Lie groups, then φ is a covering homomorphism if and only if $d\varphi : \mathfrak{g} \to \mathfrak{h}$ is an isomorphism of Lie algebras.*

Proof. If φ is a covering homomorphism, then it is a local diffeomorphism at e, hence induces a linear isomorphism of vector spaces $d\varphi : \mathfrak{g} \simeq T_eG \simeq T_eH \simeq \mathfrak{h}$. As $d\varphi$ is a Lie algebra homomorphism, it must be a Lie algebra isomorphism.

Conversely, if $d\varphi : \mathfrak{g} \to \mathfrak{h}$ is an isomorphism of Lie algebras, then φ is a local diffeomorphism everywhere. We only need to show that there exists a neighbourhood U of $e \in H$ such that $\varphi^{-1}(U)$ is a disjoint union of subsets of G, each of which is diffeomorphic to U. Indeed, $D := \ker\varphi$ is a discrete normal subgroup of G. As $G \times G \to (x, y) \mapsto xy^{-1}$ is continuous, we can find a small neighbourhood U of $e \in G$ such that $UU^{-1} \cap D = \{e\}$. Then one can easily check that $\varphi|_U$ is injective, hence $\varphi|_U$ is a diffeomorphism and

$$\varphi^{-1}(\varphi(U)) = \coprod_{g\in D} gU.$$

□

Definition 5.4. *The double (universal) covering group of* $\mathrm{SO}(n)$ *($n \geq 3$) is defined to be* ***spin group*** $\mathrm{Spin}(n)$*. The double (universal) covering group of* $\mathrm{SO}_n(\mathbb{C})$ *($n \geq 3$) is defined to be* ***complex spin group*** $\mathrm{Spin}_n(\mathbb{C})$*.*

Remark 5.5. *For indefinite signature, the group* $\mathrm{Spin}(p, q)$ *is usually discussed via Clifford algebras, which is beyond this course. It is a double cover of* $\mathrm{SO}(p, q)^0$*, and in general not simply connected. This is because even when restricted to the identity component, the fundamental group of* $\mathrm{SO}(p, q)^0$ *is in general equal to* $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$*. Thus, the universal covering group is a four-fold cover of* $\mathrm{SO}(p, q)^0$*, not a double cover. In this course, we at most discuss some low dimensional indefinite spin groups when they coincide with other classical groups. We never intend to do a serious discussion on* $\mathrm{Spin}(p, q)$ *in full generality.*

The following claims will be proved in Lecture 06 (I guess....).

Example 5.6. (1) $\mathrm{SL}_2(\mathbb{C}) \simeq \mathrm{Spin}_3(\mathbb{C})$, $\mathrm{SU}(2) \simeq \mathrm{Spin}(3)$;
(2) $\mathrm{SL}_2(\mathbb{C}) \times \mathrm{SL}_2(\mathbb{C}) \simeq \mathrm{Spin}_4(\mathbb{C})$, $\mathrm{SU}(2) \times \mathrm{SU}(2) \simeq \mathrm{Spin}(4)$;
(3) $\mathrm{Sp}_4(\mathbb{C}) \simeq \mathrm{Spin}_5(\mathbb{C})$, $\mathrm{Sp}(4) \simeq \mathrm{Spin}(5)$;
(4) $\mathrm{SL}_4(\mathbb{C}) \simeq \mathrm{Spin}_6(\mathbb{C})$, $\mathrm{SU}(4) \simeq \mathrm{Spin}(6)$.

To understand the covering homomorphism $\pi : \widetilde{G} \to G$, we need the following lemma.

Lemma 5.7 ([Sep07, Lemma 1.21])**.** *If H is a discrete normal subgroup of a connected Lie group G, then $H \leq Z_G$.*

Proof. G acts on itself by conjugation. For $h \in H$, the G-orbit generated by h is connected (as G is connected) and contained in H (as H is normal). Since H is discrete, the G-orbit generated by h must be a single point. Hence $h \in Z_G$. □

Corollary 5.8. *Let $\pi : G \to H$ be a covering homomorphism. Then* $\ker\pi$ *is discrete and contained in Z_G. Moreover, if G is the universal covering group of H, then* $\pi_1(H) \simeq \ker\pi$*.*

Proof. Obviously $\ker\pi$ is a closed normal subgroup of G, its Lie algebra is $\ker d\pi$. As $d\pi$ is a Lie algebra isomorphism, (Corollary 5.3), $\ker d\pi = 0$, hence $\ker\pi$ is discrete. This implies that $\ker\pi \leq Z(G)$ (Lemma 5.7). The last statement follows from standard topology. □

Lemma 5.9. (1) *For $G = \mathrm{SU}(n)$, $Z_G = \{\lambda \mathrm{I}_n \mid \lambda^n = 1, \lambda \in \mathbb{C}\} \simeq \mathbb{Z}/n\mathbb{Z}$.*
(2) *For $G = \mathrm{U}(n)$, $Z_G = \{\lambda \mathrm{I}_n \mid |\lambda| = 1, \lambda \in \mathbb{C}\} \simeq S^1$.*
(3) *For $G = \mathrm{Sp}(2n)$, $Z_G = \{\pm \mathrm{I}_{2n}\} \simeq \mathbb{Z}/2\mathbb{Z}$.*
(4) *For $G = \mathrm{SO}(2n + 1)$, $Z_G = \{\mathrm{I}_{2n+1}\}$.*
(5) *For $G = \mathrm{SO}(2n)$, $Z_G = \{\pm \mathrm{I}_{2n}\} \simeq \mathbb{Z}/2\mathbb{Z}$.*

Proof. The key point is to prove that the center of the above groups must be diagonal. One approach is to apply the spectral theorem of normal matrix. This approach works for unitary groups and compact symplectic groups. For orthogonal groups, we need to say more (still elementary).

Another uniform approach is to consider the standard representation of the above groups. These representations are known to be irreducible (unitary group acts transitively on unit sphere). Thus by Schur's lemma (group version), any central matrix must be scalar. □

6. Complex Structure

So far, we only use smooth structure to discuss Lie groups, though certain Lie groups such as $\mathrm{GL}_n(\mathbb{C})$ obviously enjoy more structures. We aim to discuss in this section possible complex structure on an even dimensional Lie group. As before, our main theorem will be based on a deep theorem from complex geometry, so proofs will be largely omitted.

Definition 6.1. *A **complex structure** on an even dimensional real vector space V is a linear endomorphism J of V such that $J^2 = -\mathrm{Id}$.*

Obviously, if V has a complex structure J, then V becomes a complex vector space. The complex scalar multiplication is defined via

$$(a + b\mathbf{i})v := av + bJv.$$

If V is a complex vector space, then J can be chosen to be 'multiplication by $\mathbf{i}$'.

Definition 6.2. *An **almost complex structure** on a even dimensional smooth manifold M is a smooth tensor field J of type $(1,1)$ such that at any point $p \in M$, J_p is a complex structure on the tangent space T_pM. A manifold M equipped with an almost complex structure is called **almost complex manifold**.*

Obviously, a complex manifold is almost complex. The key point is to determine when an almost complex manifold is complex.

Remark 6.3. *It is known that S^6 admits an almost complex structure. A famous open problem (at least open to the best of my knowledge) is to determine if S^6 is complex or not.*

Definition 6.4. *A **complex Lie group** is a Lie group equipped with a complex manifold structure such that group operations (i.e. multiplication and inverse) are holomorphic.*

Proposition 6.5. *A Lie group G is a complex Lie group (i.e. it can be equipped with a complex Lie group structure) if and only if there exists an **almost complex structure** on the Lie algebra $\mathfrak{g}$, i.e. at point $e \in \mathrm{G}$, there exists a complex vector space structure on $\mathfrak{g} \simeq T_e\mathrm{G}$.*

Theorem 6.6 ([Kna02, Proposition 1.110])**.**
1. *Let G be a complex Lie group. Then $\mathfrak{g}$ is a complex Lie algebra (i.e. there exists a complex structure on $\mathfrak{g}$). The exponential map is holomorphic.*
2. *Let G, H be complex Lie groups and $\varphi : G \to H$ is a smooth Lie group homomorphism. Then φ is holomorphic if and only if $d\varphi$ is complex linear.*
3. *If $\mathfrak{g}$ is a complex Lie algebra, then G admits a complex manifold structure so that G is a complex Lie group.*

Remark 6.7. *Both of the two proofs that I know require some deep theorems: one in* [Kna02, Proposition 1.110] *requires Ado's theorem (any finite-dimensional complex Lie algebra has a faithful finite-dimensional complex representation); one requires Newlander-Nirenberg theorem (an almost complex structure is integrable if and only if its Nijenhuis tensor is zero).*

Example 6.8. $\mathrm{SU}(n)$ *is never a complex Lie group. (The proof relies on the fact that* $\mathfrak{sl}_n(\mathbb{C})$ *is a simple complex Lie algebra, which will be proved later).*

For n *even, as* $\dim \mathrm{SU}(n) = n^2 - 1$ *is odd, it can never be a complex manifold. If* $\mathrm{SU}(n)$ *is a complex Lie group for some odd* n*, then* $\mathfrak{su}(n)$ *becomes a complex Lie algebra. The complexification of* $\mathfrak{su}(n)$ *is isomorphic to* $\mathfrak{sl}_n(\mathbb{C})$*, a simple complex Lie algebra. This implies that* $\mathfrak{su}(n)$ *is also simple. Yet the complexification of a complex simple Lie algebra will never be simple.*

Appendix A. Spectral Theorem of Normal Matrices

Let us recall three versions of spectral theorem of normal operators, which are usually covered in the standard linear algebra course (at least the complex case is usually covered). Recall that

Definition A.1. *Let* $F = \mathbb{R}, \mathbb{C}$ *or* $\mathbb{H}$*. A matrix* $A \in \mathrm{Mat}_n(F)$ *is said to be* ***normal*** *if* $A \cdot \overline{A}^t = \overline{A}^t \cdot A$*.*

Theorem A.2 ([Mi-To91, Chapter 1 Theorem 3.3])**.** *Let* $A \in \mathrm{Mat}_n(\mathbb{C})$*. Then the following statements are equivalent:*

(1) *A is normal;*
(2) *A is diagonalizable by a unitary matrix, i.e.* PAP^{-1} *is diagonal for some* $P \in \mathrm{U}(n)$*;*

Consider the algebra homomorphism $\theta : \mathbb{C} \to \mathrm{Mat}_2(\mathbb{R}), \quad a + b\mathbf{i} \mapsto \begin{pmatrix} a & b \\ -b & a \end{pmatrix}$.

Theorem A.3 ([Mi-To91, Chapter 1 Theorem 3.5])**.** *Let* $A \in \mathrm{Mat}_n(\mathbb{R})$*. Then the following statements are equivalent:*

(1) *A is normal;*
(2) *A is block diagonalizable by a special orthogonal matrix, i.e.* PAP^{-1} *is of the form*

$$\mathrm{diag}(r_1, r_2, \cdots, r_p, \theta(z_1), \theta(z_2), \cdots, \theta(z_q)), \quad r_j \in \mathbb{R}, z_j \in \mathbb{C}$$

for some $P \in \mathrm{SO}(n)$*;*

Theorem A.4 ([Mi-To91, Chapter 1 Theorem 3.6])**.** *Let* $A \in \mathrm{Mat}_n(\mathbb{H})$*. Then the following statements are equivalent:*

(1) *A is normal;*
(2) *A is diagonalizable by a compact symplectic matrix, i.e.* PAP^{-1} *is of the form*

$$\mathrm{diag}(z_1, z_2, \cdots, z_n), z_j \in \mathbb{H}$$

for some $P \in \mathrm{Sp}(2n)$*; moreover, all eigenvalues* z_j *are complex.*

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